

Design and Development of Bio-Mimicked Hawk Kite Ornithopter Swarm for Coastal-Mountain Environment Monitoring with BVLOS Telemetry

Institution: National Institute Of Technology Karnataka, Surathkal
Department: Department of Mechanical Engineering, NITK
Team Members:

- Chirag Satish Ingle
- Pranav Bhosale
- Thomas Johnny Bastian

Team Lead Contact: chiragingle.221me223@nitk.edu.in / +91 9356288133
Faculty Guide: Dr. Saurabh Chandrakar (Assistant Professor, Dept. Of Mechanical Engineering, NITK)

Problem Statement:	Karnataka State has ~343 km of coast running parallel to the Sahyadri mountain range, this creates a biodiverse ecosystem which supports a thriving economy of approx. \$400 billion based on fishing, agriculture, and tourism, along with major ports. This ecosystem currently faces major threats from illegal fishing, deforestation, constant flooding during monsoon and coastal flora loss. Current drone-based monitoring frameworks are insufficient to track and mitigate these sprawling threats across such complex, dual-environment terrain and varying weather conditions, leaving the region vulnerable to immense ecological and economic loss.
Objective:	• Design and Development of a Kite Hawk sized Ornithopter capable of flight range of approx. 7 kms, equipped with Camera and Infrared Sensor and onboard computation for image processing. • Programming and Deployment of Swarm Algorithms with BVLOS telemetry while maintaining Ecological Stealth monitoring with no physical or acoustic threat to local and migratory fauna.
Methodology and Approach:	This project uses a swarm of biomimetic Ornithopters to map large areas efficiently. The swarm is deployed in a "predator-prey" hierarchy, concentrating the heavy computational hardware, sensors, and navigation systems on the lead "predator" units. Hence, significantly reduces the payload requirements for the rest of the swarm. To enable remote operations, these predator units utilize GPS navigation, First-Person View (FPV) camera streams, and Beyond Visual Line of Sight (BVLOS) telemetry via 5G or SATCOM. The data collected by swarm is streamed live to the base station for real-time analysis and decision-making.

	Also, to ensure comprehensive coverage of large coastal regions, a network of low-cost, automated charging docks can be deployed to extend the swarm's operational range continuously.
Data and Tools Proposed/ Used:	• High-Resolution Photogrammetry Imagery • Low-latency video data used for BVLOS remote navigation. • QGIS / ArcGIS Pro: For compiling the collected aerial imagery, analyzing coastal geography, mapping charging dock locations, and overlaying operational ranges. • Python, Tensorflow, Pytorch, Embedded C, ROS 2, Teensy 4.1 • OpenCV & YOLO (You Only Look Once): For real-time computer vision analysis at the base station (e.g., detecting obstacles, mapping marine debris, or tracking unauthorized vessels from live FPV feed). • 5G CPE & SATCOM (e.g., Iridium/Inmarsat) For long-range, high-bandwidth transmission of telemetry and video. • LoRaWAN / Zigbee: Low-power communication protocols used locally between the "Predator - Prey" to maintain swarm formation. • Charging Dock Telemetry: Automated data regarding power consumption, battery health, & availability at docking stations.
Innovation and Novelty	• Biomimetic design enables low-altitude operations without disturbing local wildlife, unlike noisy conventional drones. • 100% 3D-printed components allow for rapid manufacturing, cost reduction, and easy fleet expansion. • Leverages natural gliding mechanics to conserve battery life and fly significantly further than standard multirotors. A novel application of Beyond Visual Line of Sight telemetry in ornithopters, enabling deep-range, remote missions.
Applications and Benefits:	• Faster detection of illegal fishing vessels and unauthorized deforestation through persistent, BVLOS aerial surveillance. • Delivery of actionable, wide-area geospatial data to local authorities to protect critical coastal infrastructure. • Silent operation among the local and migratory fauna, enables highly accurate, close-range behavioural and environmental data collection. • Rapid, post-monsoon deployment for flood mapping and damage assessment, operating reliably even in turbulent orographic winds where traditional rigid-rotor drones fail.
Future Scope:	• Scaling the regional hub model to establish a permanent, statewide environmental monitoring grid along the entire Karnataka Coastline.

	• Automate the detection and assigning the corrective measure to nearest local authority for faster threat response. • Develop a more power efficient version of ornithopter hardware to increase the flight time, range and energy consumption.
Results and Demonstrations	Figure 1: Ornithopter Prototype Figure 2: Swarm based Environment Monitoring Figure 3: Threat mitigation activity performed based on the visual data collected